

ITCS241 Database Management System

Project Phase 2

SoundCloud

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Section 1 Group 6

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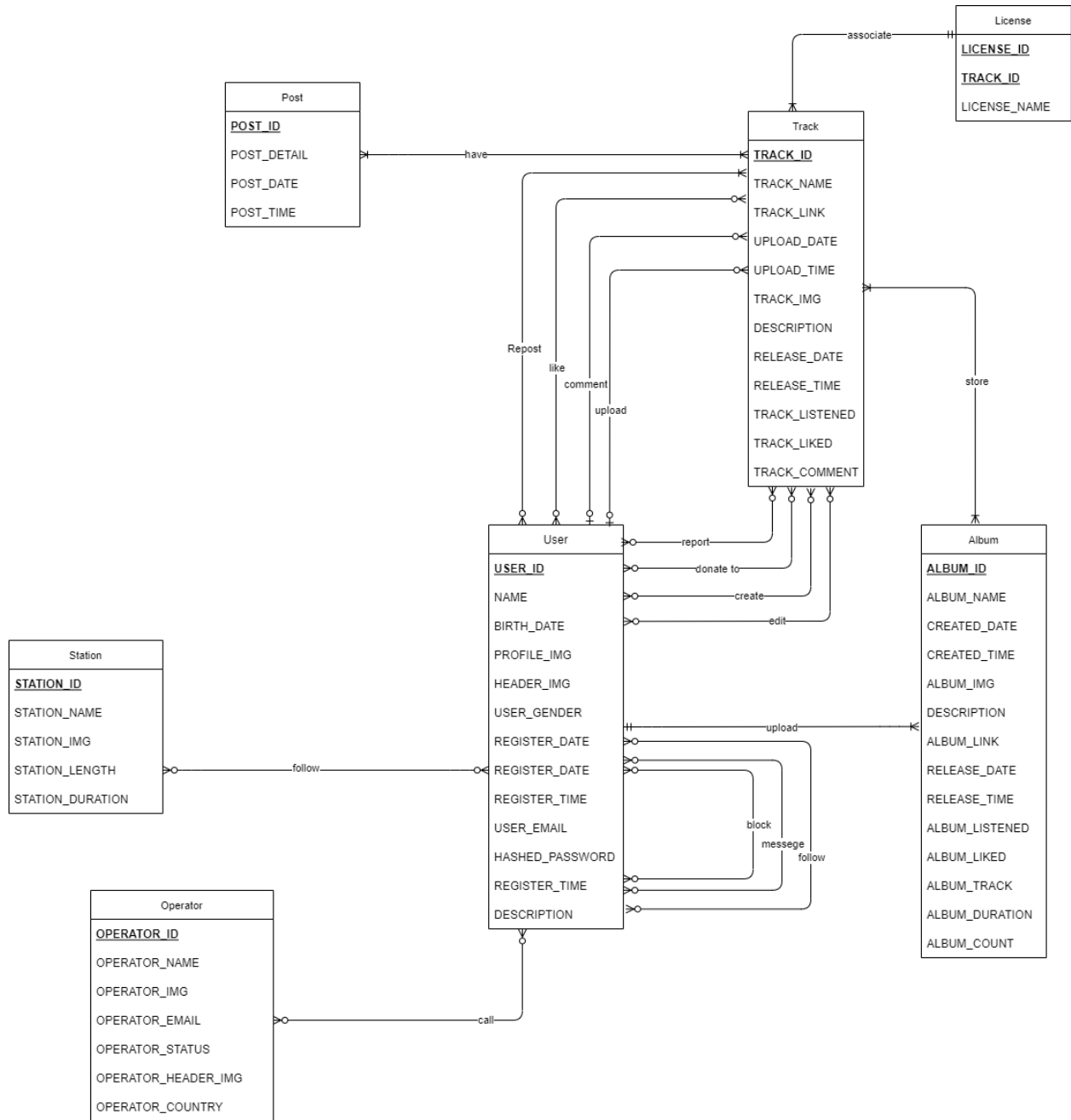
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First:

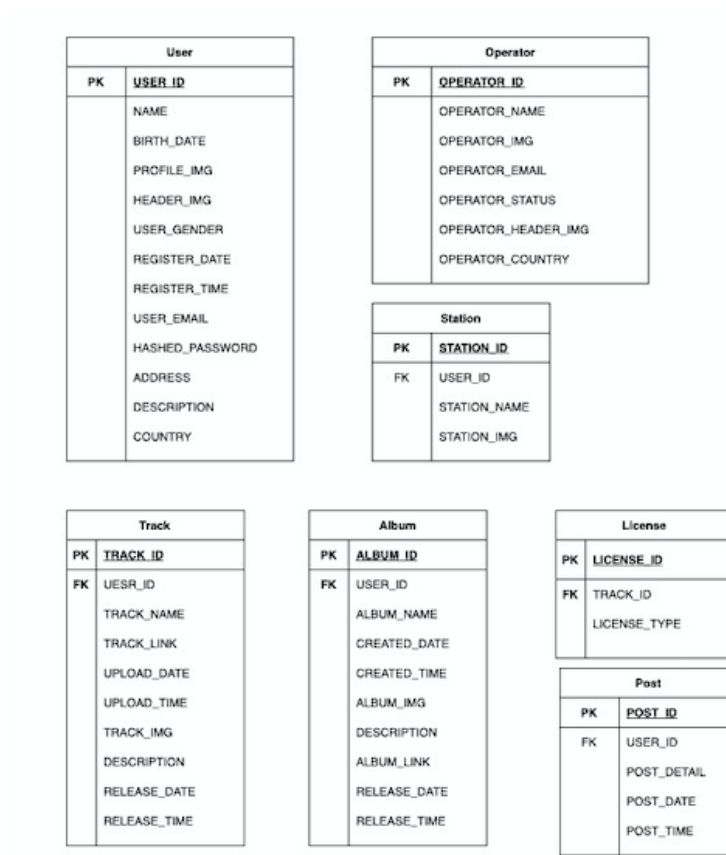
Revised ERD & Relational Schema

Revised ERD



8-steps to transform ERD to relational schema

1. Transforming regular entity types



2. Transforming weak entities

-

3. Transforming specialization or generalization

User	
PK	<u>USER_ID</u>
	NAME
	BIRTH_DATE
	PROFILE_IMG
	HEADER_IMG
	USER_GENDER
	REGISTER_DATE
	REGISTER_TIME
	USER_EMAIL
	HASHED_PASSWORD
	ADDRESS
	DESCRIPTION
	COUNTRY

Operator	
PK	<u>OPERATOR_ID</u>
	OPERATOR_NAME
	OPERATOR_IMG
	OPERATOR_EMAIL
	OPERATOR_STATUS
	OPERATOR_HEADER_IMG
	OPERATOR_COUNTRY

Track	
PK	<u>TRACK_ID</u>
FK	USER_ID
	TRACK_NAME
	TRACK_LINK
	UPLOAD_DATE
	UPLOAD_TIME
	TRACK_IMG
	DESCRIPTION
	RELEASE_DATE
	RELEASE_TIME

Album	
PK	<u>ALBUM_ID</u>
FK	USER_ID
	ALBUM_NAME
	CREATED_DATE
	CREATED_TIME
	ALBUM_IMG
	DESCRIPTION
	ALBUM_LINK
	RELEASE_DATE
	RELEASE_TIME

Post	
PK	<u>POST_ID</u>
FK	USER_ID
	POST_DETAIL
	POST_DATE
	POST_TIME

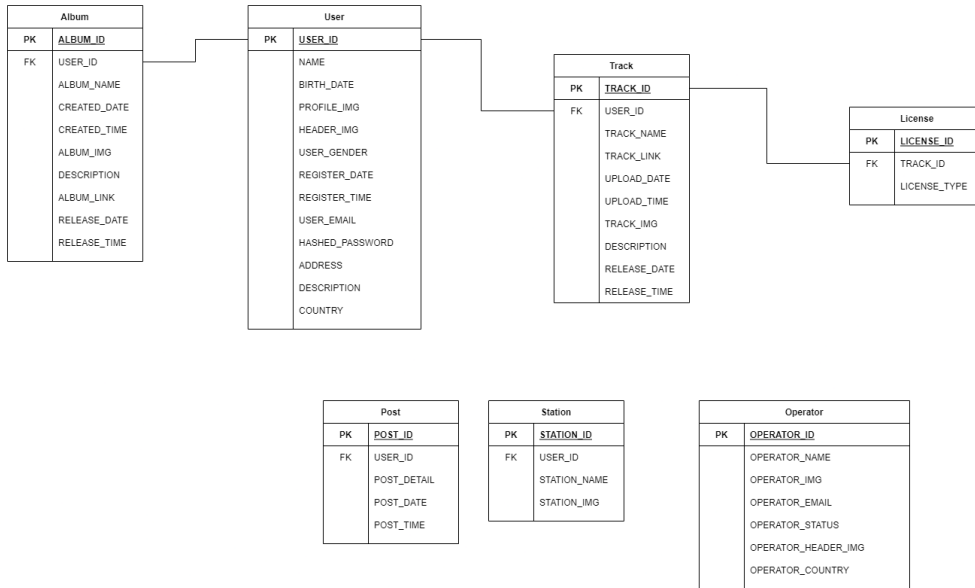
Station	
PK	<u>STATION_ID</u>
FK	USER_ID
	STATION_NAME
	STATION_IMG

License	
PK	<u>LICENSE_ID</u>
FK	TRACK_ID
	LICENSE_TYPE

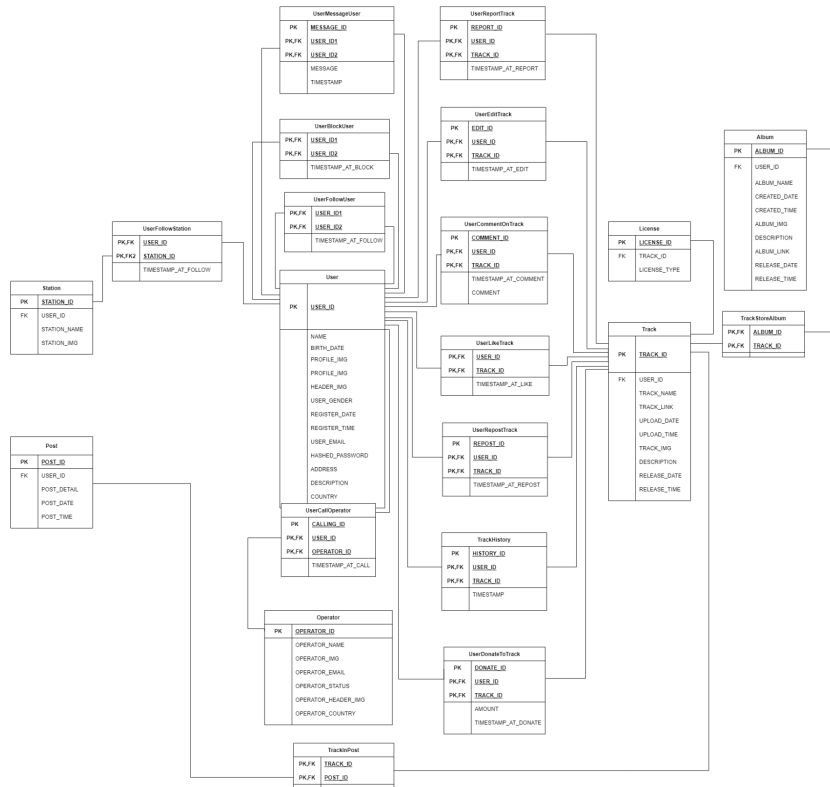
4. Transforming binary 1:1 relationship types

-

5. Transforming binary 1:N relationship types



6. Transforming binary M:N relationship types



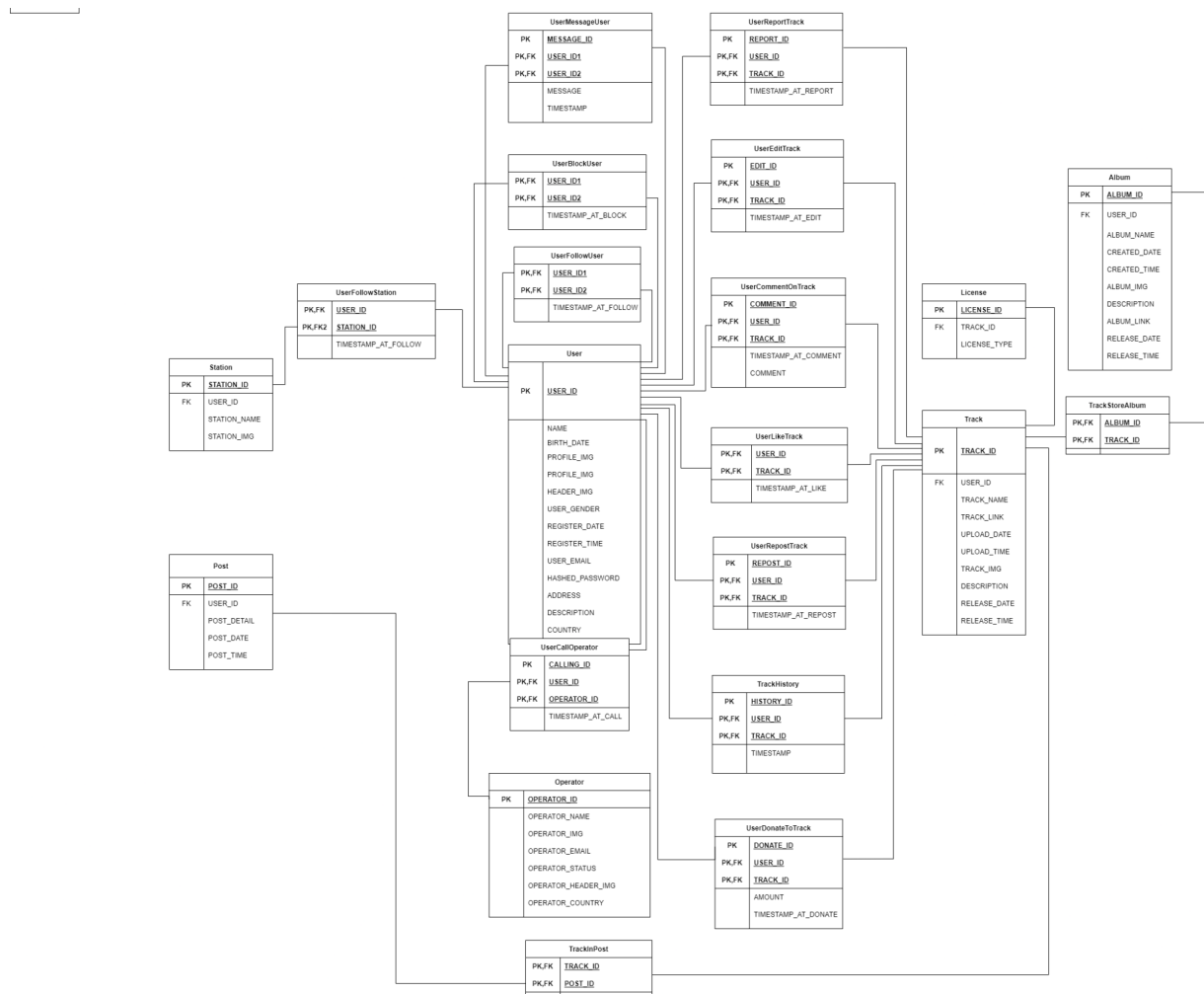
7. Transforming multi-valued attribute A in relation K

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8. Transforming n-ary relationship types

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The final relational schema



Second:

Data Dictionary

Data dictionary

Table Name	Attribute Name	Contents	Type	Format	Nullable	Range	Key	FK Referenced Table
User	USER_ID	User's ID	int				PK	
	NAME	Username	varchar(100)					
	BIRTH_DATE	User's birthdate	date	yyyy-mm-dd				
	PROFILE_IMG	User's profile image URL	varchar(500)					
	HEADER_IMG	User's header image URL	varchar(500)					
	USER_GENDER	User's gender	char(1)		Y			
	REGISTER_TIMESTAMP	User's registration timestamp	timestamp	yyyy-mm-dd hh-mi-ss				
	USER_EMAIL	User's email	varchar(50)	xxx@xxx.xxx				
	HASHED_PASSWORD	User's profile image	varchar(64)					
	ADDRESS	User's address	varchar(100)		Y			
	DESCRIPTION	User's description	varchar(500)		Y			
	COUNTRY	User's country	varchar(20)		Y			
Track	TRACK_ID	Track's ID	int	⌘			PK	
	USER_ID	Track's owner's ID	int				FK	User [USER_ID]
	TRACK_NAME	Track's name	varchar(50)					
	TRACK_LINK	Track's permalink for accessing content	varchar(100)					
	UPLOAD_TIMESTAMP	Track's upload timestamp	timestamp	yyyy-mm-dd hh-mi-ss				
	TRACK_IMG	Track's image URL	varchar(500)		Y			

	DESCRIPTION	Track's description	varchar(500)		Y			
	RELEASE_TIMESTAMP	Track's release to public timestamp	timestamp	yyyy-mm-dd hh-mi-ss	Y			
Album	ALBUM_ID	Album ID	int				PK	
	USER_ID	User's ID	int				FK	User [USER_ID]
	ALBUM_NAME	Album name	varchar(50)					
	CREATED_TIMESTAMP	Album created timestamp	timestamp					
	ALBUM_IMG	Album Image	varchar(500)					
	DESCRIPTION	Album description	varchar(500)					
	ALBUM_LINK	URL to Album	varchar(100)					
	RELEASE_TIMESTAMP	Album released timestamp	timestamp	yyyy-mm-dd hh-mi-ss	Y			
Operator	OPERATOR_ID	Operator's ID	int				PK	
	OPERATOR_NAME	Operator's Name	varchar(50)					
	OPERATOR_IMG	Operator's Image	varchar(100)					
	OPERATOR_EMAIL	Operator's E-mail	varchar(50)					
	OPERATOR_STATUS	Operator's status 0-not free, 1-free	int					
	OPERATOR_HEADER_I MG	Operator's header image	varchar(500)					
	OPERATOR_COUNTRY	Operator's located country	varchar(20)					
Post	POST_ID	Post's ID	int				PK	
	USER_ID	Post's Owner's ID	int				FK	User [USER_ID]

	POST_DETAIL	Post's text	varchar(500)					
	POST_TIMESTAMP	Posted TIMESTAMP	timestamp	yyyy-mm-dd hh-mi-ss				
Station	STATION_ID	Station's ID	int				PK	
	USER_ID	Station's Owner's ID	int				FK	User [USER_ID]
	STATION_NAME	Station's name	varchar(100)					
	STATION_IMG	Station's image URL	varchar(500)					
License	LICENSE_ID	License ID	int				PK	
	TRACK_ID	Track ID	int				FK	Track [TRACK_ID]
	LICENSE_TYPE	License name	varchar(100)					
UserMessageUser	MESSAGE_ID	Message record ID	int				PK	
	USER_ID1	First User's ID	int				PK,FK	User [USER_ID]
	USER_ID2	Second User's ID	int				PK,FK	User [USER_ID]
	MESSAGE	Message text	varchar(100)					
	TIMESTAMP	Message record timestamp	timestamp	yyyy-mm-dd hh-mi-ss				
UserBlockUser	USER_ID1	First User's ID	int				PK,FK	User [USER_ID]
	USER_ID2	Second User's ID	int				PK,FK	User [USER_ID]
	TIMESTAMP_AT_BLOCK	Blocked time	timestamp	yyyy-mm-dd hh-mi-ss				

UserFollowUser	USER_ID1	First User's ID	int				PK,FK	User [USER_ID]
	USER_ID2	Second User's ID	int				PK,FK	User [USER_ID]
	TIMESTAMP_AT_FOLLOW		timestamp	yyyy-mm-dd hh-mi-ss				
UserReportTrack	REPORT_ID	Report record ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	TIMESTAMP_AT_REPORT	Reported time	timestamp	yyyy-mm-dd hh-mi-ss				
UserEditTrack	EDIT_ID	Edit record ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	TIMESTAMP_AT_EDIT	Edited time	timestamp	yyyy-mm-dd hh-mi-ss				
UserCommentOnTrack	COMMENT_ID	Comment record ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	COMMENT	Comment text	varchar(500)					
	TIMESTAMP_AT_COMMENT	Commented time	timestamp	yyyy-mm-dd hh-mi-ss				
UserLikeTrack	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]

	TIMESTAMP_AT_LIKE	Liked time	timestamp	yyyy-mm-dd hh-mi-ss				
UserRepostTrack	REPOST_ID	Repost record ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	TIMESTAMP_AT_REPO ST	Reposted time	timestamp	yyyy-mm-dd hh-mi-ss				
TrackHistory	HISTORY_ID	Track history ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track's ID	int				PK,FK	Track [TRACK_ID]
	TIMESTAMP	timestamp at listening	timestamp	yyyy-mm-dd hh-mi-ss				
UserDonateToTrack	DONATE_ID	Donate record ID	int				PK	
	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	AMOUNT	Amount of money donate in usd	decimal(10, 2)	xxxxxxxxxx.xx				
	TIMESTAMP_AT_DONA TE	Donated time	timestamp	yyyy-mm-dd hh-mi-ss				
TrackInPost	TRACK_ID	Track ID	int				PK,FK	Track [TRACK_ID]
	POST_ID	Post's ID	int				PK,FK	Post[POST_ID]

UserFollowStation	USER_ID	User's ID	int				PK,FK	User [USER_ID]
	STATION_ID	Station's unique ID	int				PK,FK	Station[STATION_ID]
	TIMESTAMP_AT_FOLLOW	timestamp at following time	timestamp	yyyy-mm-dd hh-mi-ss				
TrackStoreAlbum	ALBUM_ID	Album's ID	int				PK,FK	Album[ALBUM_ID]
	TRACK_ID	Track's ID	int				PK,FK	Track [TRACK_ID]
UserCallOperator	CALLING_ID	Calling's unique ID	int				PK	
	USER_ID	User who contact operator's ID	int				PK,FK	User [USER_ID]
	OPERATOR_ID	Operator who got contacted's ID	int				PK,FK	Operator [OPERATOR_ID]
	TIMESTAMP_AT_CALL	timestamp at calling time	timestamp					

The sample data(eg.5-10 rows) of each table

1. Track Table :

Table: **track**

Columns:

TRACK_ID int PK
USER_ID int
 TRACK_NAME varchar(50)
 TRACK_LINK varchar(100)
 UPLOAD_TIMESTAMP timestamp
 TRACK_IMG varchar(100)
 DESCRIPTION varchar(500)
 RELEASE_TIMESTAMP timestamp

SELECT * FROM soundcloud_gr_6_sec1.track;

	TRACK_ID	USER_ID	TRACK_NAME	TRACK_LINK	UPLOAD_TIMESTAMP	TRACK_IMG
▶	1	5	chase	https://music.youtube.com/watch?v=9K2hHsT...	2016-07-27 00:00:00	https://img.hmv.co.jp/image/jacket/800/71
	2	2	Black Out	https://music.youtube.com/watch?v=MKtFvWL...	2021-11-17 00:00:00	https://i.scdn.co/image/ab67616d0000b27
	3	1	Diamond City Lights	https://music.youtube.com/watch?v=MWKw-he...	2021-05-16 00:00:00	https://images.genius.com/075a471700d8b
	4	1	One	https://music.youtube.com/watch?v=7_n_yo_9...	2017-09-18 00:00:00	https://p2.music.126.net/ZA8vgr_U_EBm4F
	5	1	Gracenote	https://music.youtube.com/watch?v=quTQs77...	2021-07-04 00:00:00	https://eimusicos.com/wp-content/uploads/2
	6	1	Last Stardust	https://music.youtube.com/watch?v=9OG04ez...	2015-06-26 00:00:00	https://www.littleoslo.com/lyj/home/wp-cor
	7	5	Big Iron	https://music.youtube.com/watch?v=zzICMIuS...	2015-05-26 00:00:00	https://images.genius.com/ec085ee0e6aa3
	8	5	March of Time	https://music.youtube.com/watch?v=aGzpeezh...	2017-05-03 00:00:00	https://s.mxmcdn.net/images-storage/albu
	9	7	Let It Be	https://music.youtube.com/watch?v=QDYfEBY...	2018-06-17 00:00:00	https://static.stereogum.com/uploads/2018
	10	7	Pretender	https://music.youtube.com/watch?v=37W7Y2R...	2019-04-16 00:00:00	https://www.lyrical-nonsense.com/wp-cont
■	NULL	NULL	NULL	NULL	NULL	NULL

2. Userrepost Track Table:

Table: **userreposttrack**

Columns:

REPOST_ID int PK
USER_ID int PK
TRACK_ID int PK
 TIMESTAMP_AT_REPOST timestamp

SELECT * FROM soundcloud_gr_6_sec1.userreposttrack;

	REPOST_ID	USER_ID	TRACK_ID	TIMESTAMP_AT_REPOST
▶	1	4	8	2021-01-14 01:41:28
	2	7	3	2021-01-22 07:23:10
	3	3	3	2021-01-27 00:04:10
	4	4	9	2021-02-25 13:55:07
	5	7	3	2021-02-28 19:03:36
	6	2	3	2021-03-07 01:03:30
	7	7	2	2021-03-07 07:39:40
	8	10	8	2021-03-10 22:32:53
	9	8	9	2021-03-13 13:32:18
	10	5	5	2021-03-27 11:18:51
	11	5	4	2021-03-28 05:01:15
	12	5	4	2021-09-04 15:36:29
	13	7	2	2021-02-05 20:42:14
	14	1	5	2021-04-05 13:54:17
	15	3	7	2021-05-18 11:14:51
	16	7	8	2021-05-26 19:12:31
	17	3	2	2021-07-02 05:12:29
	18	10	8	2021-07-29 06:14:44
■	NULL	NULL	NULL	NULL

3. Userreport track Table:

Table: **userreporttrack**

Columns:

<u>REPORT_ID</u>	int PK
<u>USER_ID</u>	int PK
<u>TRACK_ID</u>	int PK
TIMESTAMP_AT_REPORT	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.userreporttrack;
```

	REPORT_ID	USER_ID	TRACK_ID	TIMESTAMP_AT_REPORT
▶	1	5	8	2021-05-30 06:27:17
	2	4	3	2021-09-01 09:49:47
	3	1	6	2021-10-29 08:12:03
	4	8	1	2021-01-04 01:57:17
	5	10	4	2021-12-08 21:00:43
	6	4	6	2021-01-25 01:16:45
	7	1	8	2021-12-10 02:14:12
	8	2	8	2021-06-24 10:00:21
	9	9	4	2021-06-15 07:15:49
	10	3	6	2021-03-26 16:42:46
	11	8	8	2021-07-22 14:57:21
	12	9	7	2021-07-23 10:02:21
	13	1	8	2021-01-11 20:35:45
	14	1	6	2021-06-11 08:42:38
	15	10	2	2021-04-09 15:34:39
	16	9	2	2021-07-08 22:52:49
	17	2	8	2021-07-24 19:25:42
	18	7	1	2021-06-02 01:29:52
	19	5	1	2021-07-10 12:22:27

4. Usermessageuser Table:

Table: **usermessageuser**

Columns:

<u>MESSAGE_ID</u>	int PK
<u>USER_ID1</u>	int PK
<u>USER_ID2</u>	int PK
MESSAGE	varchar(100)
TIMESTAMP	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.usermessageuser;
```

	MESSAGE_ID	USER_ID1	USER_ID2	MESSAGE	TIMESTAMP
▶	14	1	1	eaque ipsa quae ab illo inventore veritatis.	2020-10-11 21:06:32
	17	1	2	sed quia consequuntur magni dolores eos qui ra...	2021-12-05 03:12:58
	10	1	6	sunt in culpa qui officia deserunt mollit anim id e...	2020-01-08 09:05:10
	18	1	8	*sequi nesciunt	2021-05-16 08:47:25
	22	1	10	magnam aliquam quaerat voluptatem.	2021-06-15 20:29:53
	27	2	3	Et harum quidem rerum facilis est et expedita di...	2021-08-31 11:07:01
	29	3	8	optio cumque nihil impedit quo minus id quod.	2021-12-10 17:07:09
	21	4	5	sed quia non numquam eius modi tempora.	2021-08-06 10:15:06
	8	4	6	velit esse cillum dolore eu fugiat nulla pariatur.	2020-08-06 07:18:46
	3	4	10	sed do eiusmod tempor incididunt ut labore et d...	2020-02-21 18:42:42
	12	5	2	accusantium doloremque laudantium.	2020-03-09 05:25:54
	30	5	2	repudiandae sint et molestiae non recusandae.	2021-10-27 00:44:57
	6	5	3	aliquip ex ea commodo consequat.	2020-04-28 19:06:45
	23	5	4	ducimus qui blanditis praesentium voluptatum d...	2021-07-20 23:28:42
	25	5	8	*cupiditate non provident	2021-11-08 12:30:43
	7	6	4	Duis aute irure dolor in reprehenderit in voluptate.	2020-06-06 02:56:59
	4	6	7	Ut enim ad minim veniam.	2020-04-04 08:21:09
	24	6	9	corrupti quos dolores et quas molestias exceptu...	2021-07-26 10:47:16
	2	7	4	consequatur vel illum est.	2020-01-21 16:47:15

5. UserlikeTrack Table:

Table: **userliketrack**

Columns:

USER_ID

int PK

TRACK_ID

int PK

TIMESTAMP_AT_LIKE

timestamp

SELECT * FROM soundcloud_gr_6_sec1.userliketrack;

	USER_ID	TRACK_ID	TIMESTAMP_AT_LIKE
▶	1	1	2021-11-04 20:47:44
	1	3	2021-10-25 20:43:30
	1	5	2021-06-29 16:41:59
	2	3	2021-05-21 11:22:04
	2	6	2021-09-06 17:38:46
	2	7	2021-11-03 15:38:32
	2	8	2021-04-13 09:26:34
	3	4	2021-11-05 03:47:42
	4	1	2021-11-11 12:51:52
	4	2	2021-06-21 04:18:22
	4	5	2021-02-18 20:18:14
	4	7	2021-08-27 19:09:54
	4	10	2021-02-18 16:50:22
	5	1	2021-09-30 04:18:36
	5	2	2021-05-31 10:59:38
	6	2	2021-07-15 08:53:47
	6	3	2021-03-29 17:10:30
	6	7	2021-09-21 20:39:12
	7	2	2021-07-06 04:04:10

6. Userfollowuser table:

Table: **userfollowuser**

Columns:

USER_ID1

int PK

USER_ID2

int PK

TIMESTAMP_AT_FOLLOW

timestamp

SELECT * FROM soundcloud_gr_6_sec1.userfollowuser;

	USER_ID1	USER_ID2	TIMESTAMP_AT_FOLLOW
▶	1	2	2021-03-04 08:22:26
	1	3	2021-06-03 18:17:43
	1	7	2021-09-28 21:25:54
	2	3	2020-03-19 10:15:06
	2	5	2021-04-29 23:46:03
	2	8	2021-10-22 02:17:11
	3	7	2021-04-20 08:54:11
	3	9	2021-05-02 11:52:05
	4	1	2021-03-16 23:32:04
	4	9	2021-09-09 14:51:34
	5	1	2021-01-05 14:15:55
	5	2	2021-02-02 18:53:36
	5	7	2021-07-30 08:53:08
	5	8	2021-04-20 20:05:22
	6	3	2021-09-13 13:46:30
	6	5	2021-06-16 09:42:41
	6	7	2021-08-30 10:05:28
	6	8	2021-07-02 15:43:26
	7	10	2021-01-16 22:54:10

7. Userfollowstation Table:

Table: **userfollowstation**

Columns:

USER_ID

int PK

STATION_ID

int PK

TIMESTAMP_AT_FOLLOW

timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.userfollowstation;
```

	USER_ID	STATION_ID	TIMESTAMP_AT_FOLLOW
▶	1	4	2021-03-14 03:49:38
	1	5	2021-02-01 23:02:38
	1	7	2021-02-16 01:01:16
	2	1	2021-01-14 06:30:53
	2	3	2021-10-30 05:33:46
	3	1	2021-03-02 16:46:46
	3	3	2021-06-08 06:07:33
	3	5	2021-10-16 16:15:45
	3	7	2021-06-05 23:11:14
	3	9	2021-02-11 21:20:22
	3	10	2021-04-02 18:03:41
	4	2	2021-08-11 19:53:31
	4	5	2021-05-10 04:17:45
	4	7	2021-11-18 09:12:46
	5	5	2021-05-25 09:01:57
	5	7	2021-08-13 12:55:52
	6	5	2021-06-05 04:26:18
	6	8	2021-06-11 02:32:30

8. Useredittrack Table:

Table: **useredittrack**

Columns:

EDIT_ID

int PK

USER_ID

int PK

TRACK_ID

int PK

TIMESTAMP_AT_EDIT

timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.useredittrack;
```

	EDIT_ID	USER_ID	TRACK_ID	TIMESTAMP_AT_EDIT
▶	1	6	7	2021-05-16 03:41:07
	2	5	3	2021-02-14 00:44:49
	3	1	3	2021-05-11 02:07:30
	4	2	8	2021-03-24 18:24:00
	5	1	5	2021-06-16 19:52:47
	6	1	2	2021-01-02 03:31:04
	7	4	4	2021-08-14 14:29:39
	8	3	7	2021-10-21 00:46:08
	9	9	8	2021-03-19 08:14:59
	10	4	10	2021-08-11 02:50:15
	11	5	4	2021-06-09 13:42:45
	12	5	2	2021-01-07 09:01:42
	13	6	5	2021-03-28 08:56:53
	14	5	4	2021-07-30 19:15:57
	15	2	8	2021-09-14 14:12:40
	16	9	7	2021-03-22 16:39:13
	17	5	3	2021-05-09 00:58:03
	18	8	1	2021-09-08 17:51:50

9. Userdonatetotrack Table:

Table: **userdonatetotrack**

Columns:

<u>DONATE_ID</u>	int PK
<u>USER_ID</u>	int PK
<u>TRACK_ID</u>	int PK
AMOUNT	decimal(10,2)
TIMESTAMP_AT_DONATE	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.userdonatetotrack;
```

	DONATE_ID	USER_ID	TRACK_ID	AMOUNT	TIMESTAMP_AT_DONATE
▶	1	10	8	110.00	2021-04-20 09:29:14
	2	3	10	60.00	2021-01-14 13:52:39
	3	9	9	50.00	2021-06-16 03:14:27
	4	6	3	150.00	2021-06-03 03:39:51
	5	9	8	110.00	2021-11-03 19:59:56
	6	4	7	130.00	2021-07-03 03:49:26
	7	10	5	30.00	2021-01-11 13:30:30
	8	3	5	60.00	2021-11-09 02:00:53
	9	7	3	30.00	2021-02-24 07:25:23
	10	5	1	140.00	2021-03-07 05:51:32
	11	1	3	80.00	2021-01-21 00:50:43
	12	7	7	30.00	2021-01-01 18:13:32
	13	1	2	100.00	2021-10-31 15:55:19
	14	5	6	70.00	2021-01-12 01:50:52
	15	4	6	130.00	2021-10-20 03:00:44
	16	4	5	20.00	2021-05-17 16:36:00
	17	10	5	120.00	2021-11-18 02:36:04
	18	8	2	40.00	2021-07-02 10:43:47
	19	10	4	30.00	2021-09-10 07:10:36

10. Usercommentontrack Table:

Table: **usercommentontrack**

Columns:

<u>COMMENT_ID</u>	int PK
<u>USER_ID</u>	int PK
<u>TRACK_ID</u>	int PK
COMMENT	varchar(500)
TIMESTAMP_AT_COMMENT	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.usercommentontrack;
```

	COMMENT_ID	USER_ID	TRACK_ID	COMMENT	TIMESTAMP_AT_COMMENT
▶	1	7	9	She says she has the ability to hear the soundtr...	2021-08-10 22:53:02
	2	5	3	He walked into the basement with the horror m...	2021-02-10 17:09:19
	3	6	8	The furnace repairman indicated the heating sy...	2021-07-06 13:23:23
	4	5	5	He waited for the stop sign to turn to a go sign.	2021-01-19 01:40:03
	5	5	8	"David subscribes to the ""stuff your tent into t...	2021-06-27 02:26:05
	6	6	1	He said he was not there yesterday, however	2021-08-09 12:13:50
	7	7	10	Blue sounded too cold at the time and yet it see...	2021-01-09 15:22:13
	8	2	9	That was how he came to win \$1 million.	2021-06-09 10:23:17
	9	8	3	Potato wedges probably are not best for relatio...	2021-03-05 19:03:45
	10	4	4	The toddler's endless tantrum caused the entire...	2021-11-16 05:14:28
	11	7	6	The random sentence generator generated a ra...	2021-08-07 06:14:28
	12	7	7	They throw cabbage that turns your brain into ...	2021-07-25 19:33:22
	13	2	4	Doris enjoyed tapping her nails on the table to ...	2021-09-29 22:30:51
	14	10	1	Mary plays the piano.	2021-05-21 16:23:24
	15	2	2	She saw the brake lights	2021-01-15 14:25:58
	16	8	8	He put heat on the wound to see what would gr...	2021-12-01 10:35:30
	17	3	4	The beach was crowded with snow leopards.	2021-06-16 19:58:59
	18	2	6	When I was little I had a car door slammed shut...	2021-03-19 03:35:12
	19	7	7	There are some fun, fun, funny, hilarious, an...	2021-09-24 07:01:55

11. UserCallOperator Table:

Table: **usercalloperator**

Columns:

<u>CALLING_ID</u>	int PK
<u>USER_ID</u>	int PK
<u>OPERATOR_ID</u>	int PK
TIMESTAMP_AT_CALL	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.UserCallOperator;
```

	CALLING_ID	USER_ID	OPERATOR_ID	TIMESTAMP_AT_CALL
▶	1	8	9	2021-07-03 12:38:16
	2	10	10	2021-09-25 02:04:49
	3	7	9	2021-07-31 06:53:51
	4	6	1	2021-01-01 12:27:16
	5	3	7	2021-06-20 09:35:14
	6	5	3	2021-03-14 06:35:49
	7	2	9	2021-09-21 04:28:26
	8	10	6	2021-02-13 07:12:12
	9	5	3	2021-09-07 08:53:18
	10	5	1	2021-07-18 05:17:35
	11	10	3	2021-11-13 04:26:04
	12	6	7	2021-07-03 04:13:26
	13	5	7	2021-05-12 02:34:16
	14	10	10	2021-10-18 01:00:19
	15	9	7	2021-01-21 02:43:07
	16	3	8	2021-09-26 07:43:39
	17	10	3	2021-10-18 07:40:05
	18	10	5	2021-07-07 02:53:39

12. UserBlockUser Table:

Table: **userblockuser**

Columns:

<u>USER_ID1</u>	int PK
<u>USER_ID2</u>	int PK
TIMESTAMP_AT_BLOCK	timestamp

```
SELECT * FROM soundcloud_gr_6_sec1.UserBlockUser;
```

	USER_ID1	USER_ID2	TIMESTAMP_AT_BLOCK
▶	1	5	2021-06-03 18:17:43
	1	8	2021-10-22 02:17:11
	2	1	2021-02-02 18:53:36
	2	3	2021-04-20 08:54:11
	2	4	2021-07-30 08:53:08
	2	10	2021-08-15 01:24:39
	3	2	2021-04-29 23:46:03
	3	5	2021-04-05 05:53:50
	4	2	2021-03-19 10:15:06
	4	8	2021-02-07 15:43:26
	5	6	2021-02-15 11:52:05
	6	3	2021-09-28 15:46:09
	6	9	2021-06-16 09:42:41
	6	10	2021-04-20 20:05:22
	7	1	2021-12-01 22:54:19
	7	2	2021-09-09 14:51:34
	7	3	2021-05-29 19:20:44
	7	4	2021-02-18 08:20:12

13. User Table:

Table: **user**

Columns:

<u>USER_ID</u>	int PK
NAME	varchar(100)
BIRTH_DATE	date
PROFILE_IMG	varchar(500)
HEADER_IMG	varchar(500)
USER_GENDER	char(1)
REGISTER_TIMESTAMP	timestamp
USER_EMAIL	varchar(50)
HASHED_PASSWORD	varchar(64)
ADDRESS	varchar(100)
DESCRIPTION	varchar(500)
COUNTRY	varchar(20)

SELECT * FROM soundcloud_gr_6_sec1.User;

	USER_ID	NAME	BIRTH_DATE	PROFILE_IMG	HEADER_IMG	USER_GENDER
▶	1	Rujiphart	2001-07-19	https://i.sndcdn.com/avatars-000436260870-...	https://media.istockphoto.com/photos/flowers-...	M
	2	Phuriwat	2002-06-05	https://st3.depositphotos.com/15648834/1793...	https://images.unsplash.com/reserve/bOvf94d...	M
	3	Bunradar	2000-03-31	https://image.shutterstock.com/image-vector/f...	https://assets.entrepreneur.com/content/3x2/...	F
	4	Waranat	2000-01-11	https://cucans.in.th/wp-content/themes/cera/a...	https://think.kera.org/wp-content/uploads/202...	M
	5	Kanyanut	1999-09-08	https://i.redd.it/v0caqchb741.jpg	https://encrypted-tbn0.gstatic.com/images?q=...	F
	6	Padon	1999-08-10	https://1.bp.blogspot.com/-Muc21xaQJYg/Xduh...	https://encrypted-tbn0.gstatic.com/images?q=...	M
	7	Pongpop	2001-05-10	https://wallpapercave.com/wp/wp8953919.gif	https://www.cantifix.co.uk/globalassets/natura...	M
	8	Kanin	2001-05-03	https://telegram.org/file/464001801/4/pPObBO...	https://www.goturkeytourism.com/assets/img/t...	M
	9	Watsapol	1999-11-18	https://images.unsplash.com/photo-152966525...	https://www.ffa.org/wp-content/uploads/2018...	M
	10	Paveena	2000-10-02	https://encrypted-tbn0.gstatic.com/images?q=...	https://cdn-a.william-reed.com/var/vrbm_gb_f...	F
*	NULL	NULL	NULL	NULL	NULL	NULL

14. TrackStoreAlbum Table:

Table: **trackstorealbum**

Columns:

<u>ALBUM_ID</u>	int PK
<u>TRACK_ID</u>	int PK

SELECT * FROM soundcloud_gr_6_sec1.TrackStoreAlbum;

	ALBUM_ID	TRACK_ID
▶	10	1
	5	2
	6	2
	1	3
	4	3
	5	3
	1	4
	3	4
	8	4
	10	4
	3	5
	6	5
	8	5
	9	5
	5	6
	10	6
	1	8
	3	8
*	NULL	NULL

15. TrackInPost Table:

Table: **trackinpost**

Columns:

<u>TRACK_ID</u>	int PK
<u>POST_ID</u>	int PK

SELECT * FROM soundcloud_gr6_sec_1.TrackInPost;

	TRACK_ID	POST_ID
▶	4	1
	6	1
	7	1
	9	1
	1	2
	5	3
	10	3
	1	4
	5	4
	1	5
	5	5
	6	5
	9	5
	3	6
	5	6
	7	6
	9	6
	3	7
	7	7

16. TrackHistory Table:

Table: **trackhistory**

Columns:

<u>HISTORY_ID</u>	int PK
<u>USER_ID</u>	int PK
<u>TRACK_ID</u>	int PK
TIMESTAMP	timestamp

SELECT * FROM soundcloud_gr6_sec_1.TrackHistory;

	HISTORY_ID	USER_ID	TRACK_ID	TIMESTAMP
▶	1	1	2	2021-10-26 04:45:55
	2	8	2	2021-09-08 06:53:39
	3	6	4	2021-08-26 05:26:59
	4	4	10	2021-06-03 12:20:41
	5	7	7	2021-03-07 23:23:38
	6	9	3	2021-09-04 07:51:39
	7	2	2	2021-03-23 05:14:18
	8	5	4	2021-11-14 22:14:26
	9	5	8	2021-04-08 15:38:58
	10	2	8	2021-02-25 19:13:40
	11	5	3	2021-08-15 03:37:49
	12	5	5	2021-02-03 13:26:56
	13	6	9	2021-05-21 07:50:34
	14	7	3	2021-04-30 17:59:36
	15	3	3	2021-01-02 05:35:17
	16	9	6	2021-03-12 17:00:58
	17	5	9	2021-02-13 09:18:38
	18	6	5	2021-07-23 19:05:28
	19	4	7	2021-03-20 15:16:26

17. Station Table:

Table: **station**

Columns:

<u>STATION_ID</u>	int PK
<u>USER_ID</u>	int
STATION_NAME	varchar(100)
STATION_IMG	varchar(500)

```
SELECT * FROM soundcloud_gr6_sec_1.Station;
```

	STATION_ID	USER_ID	STATION_NAME	STATION_IMG
▶	1	3	Onlikiszz	https://i1.sndcdn.com/avatars-000336917074-...
	2	5	Voccaloider	https://i1.sndcdn.com/avatars-000134074600-...
	3	10	Bookiezz	https://i1.sndcdn.com/avatars-000198480382-...
	4	8	Jeanne-MuNe	https://i1.sndcdn.com/avatars-000103969207-...
	5	6	Lament47	https://i1.sndcdn.com/avatars-000449974293-...
	6	5	Jyu / Just Q	https://i1.sndcdn.com/avatars-czxGomu1exwY...
	7	9	Ki	https://i1.sndcdn.com/avatars-gxzFkZPhRj9gRi...
	8	7	urata	https://i1.sndcdn.com/avatars-g67zyuNPT3lF2l...
	9	1	sora lee	https://i1.sndcdn.com/avatars-a6iK0RJSTvZyA...
	10	2	Moggy3.0	https://i1.sndcdn.com/avatars-D7IbuD1oAtZW...
•	NULL	NULL	NULL	NULL

18. Post Table:

Table: **post**

Columns:

<u>POST_ID</u>	int PK
<u>USER_ID</u>	int
POST_DETAIL	varchar(500)
POST_TIMESTAMP	timestamp

```
SELECT * FROM soundcloud_gr6_sec_1.Post;
```

	POST_ID	USER_ID	POST_DETAIL	POST_TIMESTAMP
▶	1	8	So long and thanks for the fish.	2021-04-18 22:10:37
	2	8	The random sentence generator generated a ra...	2021-04-29 17:29:18
	3	2	At that moment he wasn't listening to music	2021-06-03 17:21:44
	4	6	Now I need to ponder my existence and ask my...	2021-02-07 16:03:13
	5	10	As time wore on	2021-09-08 03:51:16
	6	8	She is never happy until she finds something to ...	2021-10-14 01:22:32
	7	6	With the high wind warning	2021-11-17 18:05:58
	8	6	He figured a few sticks of dynamite were easier...	2021-08-10 00:52:59
	9	7	Even with the snow falling outside	2021-08-20 07:16:06
	10	10	There aren't enough towels in the world to stop...	2021-01-01 00:36:07
	11	6	Baby wipes are made of chocolate stardust.	2021-02-28 11:00:56
	12	6	The sky is clear; the stars are twinkling.	2021-01-04 02:17:07
	13	6	The child's favorite Christmas gift was the large ...	2021-01-02 00:11:44
	14	10	You have every right to be angry	2021-06-11 17:27:01
	15	7	Charles ate the french fries knowing they would...	2021-05-30 02:25:12
	16	2	100 years old is such a young age if you happe...	2021-03-27 07:14:53
	17	6	When confronted with a rotary dial phone the t...	2021-03-26 07:44:51
	18	6	He decided that the time had come to be strong...	2021-09-19 04:18:19
	19	1	Breakfast butter and jelly caused the slides to slide...	2021-09-01 13:15:41

19. Operator Table:

Table: **operator**

Columns:

OPERATOR_ID	int PK
OPERATOR_NAME	varchar(50)
OPERATOR_IMG	varchar(500)
OPERATOR_EMAIL	varchar(50)
OPERATOR_STATUS	int
OPERATOR_HEADER_IMG	varchar(500)
OPERATOR_COUNTRY	varchar(20)

SELECT * FROM soundcloud_gr6_sec_1.Operator;

	OPERATOR_ID	OPERATOR_NAME	OPERATOR_IMG	OPERATOR_EMAIL	OPERATOR_STATUS	OPERATOR_HEADE
▶	1	Akara	https://www.ict.mahidol.ac.th/wp-content/uplo...	akara.sup@mahidol.edu	0	https://wallup.net/v
	2	Pawita	https://www.ict.mahidol.ac.th/wp-content/uplo...	pawitra.lia@mahidol.ac.th	0	https://www.walpa
	3	Siripen	https://www.ict.mahidol.ac.th/wp-content/uplo...	siripen.pon@mahidol.ac.th	0	https://www.pcdear
	4	Boonsit	https://www.ict.mahidol.ac.th/wp-content/uplo...	boonsit.yimmahidol.ac.th	0	https://www.walpa
	5	Pilaluck	https://www.ict.mahidol.ac.th/wp-content/uplo...	pilaluck.panmahidol.ac.th	1	https://www.walpa
	6	Wudhichart	https://www.ict.mahidol.ac.th/wp-content/uplo...	wudhichart.sawmahidol.ac.th	1	https://www.walpa
	7	Ittipon	https://www.ict.mahidol.ac.th/wp-content/uplo...	ittipon.rasmahidol.ac.th	0	https://www.walpa
	8	Snit	https://www.ict.mahidol.ac.th/wp-content/uplo...	snit.san@mahidol.ac.th	0	https://www.walpa
	9	Jidapa	https://www.ict.mahidol.ac.th/wp-content/uplo...	jidapa.kramahidol.ac.th	1	https://www.walpa
	10	Ananta	https://www.ict.mahidol.ac.th/wp-content/uplo...	ananta.sri@mahidol.edu	0	https://www.walpa
•	NULL	NULL	NULL	NULL	NULL	NULL

20. License Table:

Table: **license**

Columns:

LICENSE_ID	int PK
TRACK_ID	int
LICENSE_TYPE	varchar(100)

SELECT * FROM soundcloud_gr6_sec_1.License;

	LICENSE_ID	TRACK_ID	LICENSE_TYPE
▶	1	1	Public Performing Right
	2	2	Public Performing Right
	3	3	Digital Performance Right in Sound Recordings
	4	4	Synchronization License
	5	5	Synchronization License
	6	6	Reproduction Right
	7	7	Reproduction Right
	8	8	Reproduction Right
	9	9	Public Performing Right
	10	10	Synchronization License
•	NULL	NULL	NULL

21. Album Table:

Table: **album**

Columns:

ALBUM_ID	int PK
USER_ID	int
ALBUM_NAME	varchar(100)
CREATED_TIMESTAMP	timestamp
ALBUM_IMG	varchar(100)
DESCRIPTION	varchar(500)
ALBUM_LINK	varchar(100)
RELEASE_TIMESTAMP	timestamp

SELECT * FROM soundcloud_gr6_sec_1.Album;

	ALBUM_ID	USER_ID	ALBUM_NAME	CREATED_TIMESTAMP	ALBUM_IMG	DESCRIPTION
	1	5	Formula of Love: O+T=<3	2021-08-07 06:00:01	https://i1.sndcdn.com/artworks-3b0518b8-77e...	My lovely alb
	2	2	The 1st	2021-11-16 10:30:50	https://i1.sndcdn.com/artworks-6b1c63fb-8d73...	He is 10 cm b
	3	1	TAEYEON - 불타 (Spark) [Purpose - The 2nd Alb...	2016-07-09 09:13:35	https://i1.sndcdn.com/artworks-000639061858...	Ya e ya e ya
	4	7	INSPIRED	2018-01-10 10:18:20	https://i1.sndcdn.com/artworks-000006721749...	Vi tee duu mi
	5	9	Album Sampler Out Now	2017-10-23 06:40:41	https://i1.sndcdn.com/artworks-000010417501...	"I found the l
	6	3	THE ALBUM	2015-04-19 19:34:12	https://i1.sndcdn.com/artworks-6c8141af-033c...	Boombayah y
	7	5	The Feels	2021-06-07 10:00:57	https://i1.sndcdn.com/artworks-d774b3af-23d1...	I got the feel
	8	4	아이유 (IU) - Love poem (5th Mini Album) (Blue...	2021-08-14 17:15:25	https://i1.sndcdn.com/artworks-000639202423...	I luv U
	9	7	FAKE LOVE / Airplane pt.2	2018-03-03 13:26:39	https://i1.sndcdn.com/artworks-DkCfhqOcdN0...	Bangkok Trai
	10	8	ITZY (있지) - GUESS WHO 'MAFIA 마, 피, 아. In t...	2021-05-29 13:04:51	https://i1.sndcdn.com/artworks-LH2xkpQnITi1c...	I'm a M.A.F.I
»	NULL	NULL	NULL	NULL	NULL	NULL

Basic queries

1. Find the user who lives in Thailand

```
SELECT USER_ID, NAME, PROFILE_IMG
FROM User
WHERE COUNTRY = 'TH'
ORDER BY USER_ID ASC;
```

	USER_ID	NAME	PROFILE_IMG
▶	1	Rujiphart	https://i1.sndcdn.com/avatars-000436260870-...
*	NULL	NULL	NULL

2. Find the track which begins with 'B'

```
SELECT TRACK_NAME, TRACK_LINK, TRACK_IMG
FROM Track
WHERE TRACK_NAME LIKE 'B%'
ORDER BY TRACK_ID ASC;
```

	TRACK_NAME	TRACK_LINK	TRACK_IMG
▶	Black Out	https://music.youtube.com/watch?v=MktFvWL...	https://i.scdn.co/image/ab67616d0000b273f0f...
	Big Iron	https://music.youtube.com/watch?v=zzICMIu5...	https://images.genius.com/ec085ee0e6aa35e6...

3. Find the album which created by the same user by ID, such as '7'

```
SELECT ALBUM_NAME, ALBUM_IMG, ALBUM_LINK, DESCRIPTION
FROM Album
WHERE USER_ID = 7
ORDER BY ALBUM_ID;
```

	ALBUM_NAME	ALBUM_IMG	ALBUM_LINK	DESCRIPTION
▶	INSPIRED	https://i1.sndcdn.com/artworks-000006721749...	https://soundcloud.com/polycat/sets/inspired-2	Vi tee duu mi yak a rai.
	FAKE LOVE / Airplane pt.2	https://i1.sndcdn.com/artworks-DkCfhqOcdN0...	https://soundcloud.com/bangtan/sets/fake-lov...	Bangkok Train Station.

4. Find the operator who is available by STATUS (1 = available)

```
SELECT OPERATOR_NAME, OPERATOR_IMG, OPERATOR_EMAIL
FROM Operator
WHERE OPERATOR_STATUS = 1
ORDER BY OPERATOR_ID;
```

	OPERATOR_NAME	OPERATOR_IMG	OPERATOR_EMAIL
▶	Pililuck	https://www.ict.mahidol.ac.th/wp-content/uplo...	pilailuck.panmahidol.ac.th
	Wudhichart	https://www.ict.mahidol.ac.th/wp-content/uplo...	wudhichart.sawmahidol.ac.th
	Jidapa	https://www.ict.mahidol.ac.th/wp-content/uplo...	jidapa.kramahidol.ac.th

5. Show the users' post details who have the ID less than 6

```
SELECT USER_ID, POST_DETAIL, POST_TIMESTAMP
FROM Post
WHERE USER_ID <=5
ORDER BY USER_ID;
```

	USER_ID	POST_DETAIL	POST_TIMESTAMP
▶	1	Peanut butter and jelly caused the elderly lady ...	2021-08-01 12:15:41
	2	At that moment he wasn't listening to music	2021-06-03 17:21:44
	2	100 years old is such a young age if you happe...	2021-03-27 07:14:53
	2	They looked up at the sky and saw a million stars.	2021-05-18 06:49:57
	2	At that moment I was the most fearsome weas...	2021-01-24 07:26:34
	2	Everyone says they love nature until they realiz...	2021-02-24 18:37:09
	5	The bullet pierced the window shattering it befo...	2021-11-12 10:02:12
	5	In that instant	2021-03-30 21:43:50

6. Number of people who use gmail

```
SELECT COUNT(*) AS "number of people who use gmail"
FROM User
WHERE USER_EMAIL LIKE "%@gmail.com"
```

	number of people who use gmail
▶	5

7. Users which have the email pattern "@gmail.com" or "@hotmail.com"

```
SELECT USER_ID, NAME AS USERNAME, USER_EMAIL, "@gmail.com" AS E_MAIL_PATTERN
FROM USER
WHERE USER_EMAIL LIKE "%@gmail%"
UNION
SELECT USER_ID, NAME AS USERNAME, USER_EMAIL, "@hotmail.com" AS
E_MAIL_PATTERN
FROM USER
WHERE USER_EMAIL LIKE "%@hotmail%";
```

USER_ID	USERNAME	USER_EMAIL	E_MAIL_PATTERN
1	Rujiphart	Rujiphartzaaa@gmail.com	@gmail.com
3	Bunradar	BunradarBarr@gmail.com	@gmail.com
5	Kanyanut	Kanyanuttki@gmail.com	@gmail.com
7	Pongpop	Pongpoppop@gmail.com	@gmail.com
10	Paveena	Paveenaiei@gmail.com	@gmail.com
2	Phuriwat	Phuriwathahaha@hotmail.com	@hotmail.com
4	Waranat	Waranatzz@hotmail.com	@hotmail.com
9	Watsapol	Watsapolzarzar@hotmail.com	@hotmail.com

8. Find the user's station name by user's ID, such as user's ID is 9

```
SELECT STATION_NAME, STATION_IMG
FROM Station
WHERE USER_ID = '9';
```

	STATION_NAME	STATION_IMG
▶	Ki	https://i1.sndcdn.com/avatars-gxzFkZPhRj9gRi...

9. Find the license type of the track by track's ID, such as 4

```
SELECT LICENSE_TYPE
FROM License
WHERE TRACK_ID = '4';
```

	LICENSE_TYPE
▶	Synchronization License

10. number of track that has been played since 2021/08/01 0:00:00 UTC

```
SELECT COUNT(*) AS "track played"
FROM TrackHistory th
WHERE th.TIMESTAMP > "2021/08/01 0:00:00";
```

	track played
▶	9

11. Show user's message history which ordered by date and time, such as history of user's ID is 1

```
SELECT USER_ID2, MESSAGE, timestamp
FROM UserMessageUser
WHERE USER_ID1 = 1
ORDER BY TIMESTAMP ASC;
```

	USER_ID2	MESSAGE	timestamp
▶	6	sunt in culpa qui officia deserunt mollit anim id e...	2020-01-08 09:05:10
	1	eaque ipsa quae ab illo inventore veritatis.	2020-10-11 21:06:32
	8	"sequi nesciunt	2021-05-16 08:47:25
	10	magnam aliquam quaerat voluptatem.	2021-06-15 20:29:53
	2	sed quia consequuntur magni dolores eos qui ra...	2021-12-05 03:12:58

12. Count the number of license types used

```
SELECT LICENSE_TYPE AS "License type", COUNT(TRACK_ID) AS "License count"
FROM License
GROUP BY LICENSE_TYPE;
```

	License type	License count
▶	Public Performing Right	3
	Digital Performance Right in Sound Recordings	1
	Synchronization License	3
	Reproduction Right	3

13. Find the users who are blocked by the user whose ID is 10

```
SELECT USER_ID2, TIMESTAMP_AT_BLOCK
FROM UserBlockUser
WHERE USER_ID1 = '10'
ORDER BY TIMESTAMP_AT_BLOCK ASC;
```

	USER_ID2	TIMESTAMP_AT_BLOCK
▶	5	2021-03-04 08:22:26
	1	2021-03-06 17:29:35
	4	2021-11-01 21:40:58

14. Show the users id who follow user whose id is 3

```
SELECT USER_ID2
FROM UserFollowUser
WHERE USER_ID1 = '3'
ORDER BY USER_ID2 ASC;
```

	USER_ID2
▶	7
	9

15. Show the users who like the track whose id is 2. It is ordered by date and time.

```
SELECT USER_ID, TIMESTAMP_AT_LIKE
FROM UserLikeTrack
WHERE TRACK_ID = 2
ORDER BY TIMESTAMP_AT_LIKE ASC;
```

	USER_ID	TIMESTAMP_AT_LIKE
▶	5	2021-05-31 10:59:38
	4	2021-06-21 04:18:22
	7	2021-07-02 06:04:19
	6	2021-07-15 08:53:47
	8	2021-09-17 11:42:52

16. Count operator called for each month

```
SELECT YEAR(TIMESTAMP_AT_CALL) AS "year", MONTH(TIMESTAMP_AT_CALL) AS
"month", COUNT(*) AS "operator called count"
FROM UserCallOperator
GROUP BY YEAR(TIMESTAMP_AT_CALL), MONTH(TIMESTAMP_AT_CALL)
ORDER BY TIMESTAMP_AT_CALL;
```

	year	month	operator called count
▶	2021	1	3
	2021	2	1
	2021	3	2
	2021	5	2
	2021	6	4
	2021	7	8
	2021	8	1
	2021	9	4
	2021	10	4
	2021	11	1

17. Show the amount of donation that each user donated

```
SELECT USER_ID, SUM(AMOUNT) AS AMOUNT
FROM UserDonateToTrack
GROUP BY USER_ID
ORDER BY USER_ID;
```

	USER_ID	AMOUNT
▶	1	180.00
	2	210.00
	3	230.00
	4	410.00
	5	250.00
	6	220.00
	7	60.00
	8	80.00
	9	260.00
	10	290.00

18. Show the amount of the follower of each station

```

SELECT STATION_ID, COUNT(USER_ID)
FROM UserFollowStation
GROUP BY STATION_ID
ORDER BY STATION_ID;

```

	STATION_ID	COUNT(USER_ID)
▶	1	2
	2	1
	3	3
	4	3
	5	7
	6	1
	7	4
	8	2
	9	4
	10	3

19. Count the day of registration of the user.

```

SELECT
CASE
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 1 THEN 'Sunday'
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 2 THEN 'Monday'
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 3 THEN 'Tuesday'
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 4 THEN 'Wednesday'
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 5 THEN 'Thursday'
  WHEN DAYOFWEEK(REGISTER_TIMESTAMP) = 6 THEN 'Friday'
  ELSE 'Saturday'
END AS Day_of_register,
COUNT(*) AS COUNT_OF_USER
FROM USER
GROUP BY DAYOFWEEK(REGISTER_TIMESTAMP)
ORDER BY DAYOFWEEK(REGISTER_TIMESTAMP);

```

Day_of_register	COUNT_OF_USER
Sunday	3
Tuesday	2
Wednesday	1
Thursday	2
Friday	1
Saturday	1

20. Average, std, min, max, variance of string length

```
SELECT ROUND(AVG(CHAR_LENGTH(MESSAGE)), 2) AS "average string length",  
       ROUND(STD(CHAR_LENGTH(MESSAGE)), 2) AS "std string length",  
       MIN(CHAR_LENGTH(MESSAGE)) AS "min string length",  
       MAX(CHAR_LENGTH(MESSAGE)) AS "max string length",  
       VARIANCE(CHAR_LENGTH(MESSAGE)) AS "var string lenght"  
FROM UserMessageUser;
```

	average string length	std string length	min string length	max string length	var string lenght
►	44.50	14.53	15	66	211.05

Advanced queries

1. Amount of money donated to each track.

```
SELECT t.TRACK_NAME AS "track name", SUM(udtt.AMOUNT) AS "donation amount"
FROM UserDonateToTrack udtt
INNER JOIN Track t ON udtt.TRACK_ID = t.TRACK_ID
GROUP BY t.TRACK_NAME;
```

	track name	donation amount
▶	chase	150.00
	Black Out	250.00
	Diamond City Lights	310.00
	One	160.00
	Gracernote	280.00
	Last Stardust	200.00
	Big Iron	220.00
	March of Time	470.00
	Let It Be	50.00
	Pretender	100.00

2. Count number of donations occurs for each track

```
SELECT t.TRACK_NAME AS "track name", COUNT(udtt.AMOUNT) AS "donation count"
FROM UserDonateToTrack udtt
INNER JOIN Track t ON udtt.TRACK_ID = t.TRACK_ID
GROUP BY t.TRACK_NAME;
```

	track name	donation count
▶	chase	2
	Black Out	3
	Diamond City Lights	4
	One	3
	Gracernote	5
	Last Stardust	2
	Big Iron	4
	March of Time	4
	Let It Be	1
	Pretender	2

3. Count number of follower for each user and show name

```
SELECT u.USER_ID AS "UserID", u.NAME AS "UserName", COUNT(ufu.USER_ID2) AS "follower"
FROM UserFollowUser ufu
INNER JOIN User u ON ufu.USER_ID1 = u.USER_ID
GROUP BY u.USER_ID;
```

	UserID	UserName	follower
▶	1	Rujiphart	3
	2	Phuriwat	3
	3	Bunradar	2
	4	Waranat	2
	5	Kanyanut	4
	6	Padon	5
	7	Pongpop	4
	8	Kanin	4
	9	Watsapol	1
	10	Paveena	2

4. Count number of comment on each track, sort by comment count

```
SELECT ucot.TRACK_ID, t.TRACK_NAME, COUNT(ucot.USER_ID) AS "comment count"
FROM UserCommentOnTrack ucot
INNER JOIN Track t ON ucot.TRACK_ID = t.TRACK_ID
GROUP BY ucot.TRACK_ID
ORDER BY COUNT(ucot.USER_ID) DESC;
```

	TRACK_ID	TRACK_NAME	comment count
▶	10	Pretender	5
	3	Diamond City Lights	4
	7	Big Iron	4
	8	March of Time	4
	4	One	3
	6	Last Stardust	3
	1	chase	2
	2	Black Out	2
	9	Let It Be	2
	5	Gracernote	1

5. Count number of track released by each user

```
SELECT u.USER_ID AS "UserId", u.NAME AS "Username", COUNT(TRACK_ID) AS "number of
track own"
FROM User u
LEFT JOIN Track t ON u.USER_ID = t.USER_ID
GROUP BY u.USER_ID
ORDER BY COUNT(TRACK_ID) DESC;
```

	UserId	Username	number of track own
▶	1	Rujiphart	4
	5	Kanyanut	3
	7	Pongpop	2
	2	Phuriwat	1
	3	Bunradar	0
	4	Waranat	0
	6	Padon	0
	8	Kanin	0
	9	Watsapol	0
	10	Paveena	0

6. Count number of tracks in each album

```
SELECT tsa.ALBUM_ID, a.ALBUM_NAME, COUNT(tsa.TRACK_ID) AS "number of track in album"
FROM TrackStoreAlbum tsa
INNER JOIN Album a ON tsa.ALBUM_ID = a.ALBUM_ID
GROUP BY ALBUM_ID;
```

	ALBUM_ID	ALBUM_NAME	number of track in album
▶	1	Formula of Love: O+T=<3	4
	2	The 1st	2
	3	TAEYEON - 불티 (Spark) [Purpose - The 2nd Alb...	5
	4	INSPIRED	4
	5	Album Sampler Out Now	5
	6	THE ALBUM	3
	8	아이유 (IU) - Love poem (5th Mini Album) (Blue...	2
	9	FAKE LOVE / Airplane pt.2	1
	10	ITZY (있지) - GUESS WHO 'MAFIA 마.피.아. In t...	4

7. Count listener for each track

```
SELECT th.TRACK_ID, t.TRACK_NAME, COUNT(th.USER_ID) AS "listener count"
FROM TrackHistory th
RIGHT JOIN Track t ON th.TRACK_ID = t.TRACK_ID
GROUP BY th.TRACK_ID
ORDER BY COUNT(th.USER_ID) DESC;
```

	TRACK_ID	TRACK_NAME	listener count
▶	3	Diamond City Lights	6
	2	Black Out	5
	7	Big Iron	4
	4	One	3
	6	Last Stardust	3
	5	Gracernote	2
	8	March of Time	2
	9	Let It Be	2
	10	Pretender	2
	1	chase	1

8. Count the number of times that each operator has calls.

```
SELECT O.OPERATOR_NAME, COUNT(U.OPERATOR_ID) AS OPERATION_COUNT
FROM usercalloperator U
INNER JOIN operator O ON O.OPERATOR_ID = U.OPERATOR_ID
GROUP BY U.OPERATOR_ID
ORDER BY COUNT(U.OPERATOR_ID) DESC;
```

	OPERATOR_NAME	OPERATION_COUNT
▶	Siripen	5
	Ittipon	5
	Jidapa	5
	Akara	4
	Pililuck	3
	Ananta	3
	Wudhichart	2
	Pawita	1
	Boonsit	1
	Snit	1

9. count view for each track and for each user

```
SELECT u.NAME, COUNT(th.TRACK_ID) AS "listener count"
FROM User u
LEFT JOIN Track t ON u.USER_ID = t.USER_ID
LEFT JOIN TrackHistory th ON t.TRACK_ID = th.TRACK_ID
GROUP BY u.USER_ID
ORDER BY COUNT(th.TRACK_ID) DESC;
```

	NAME	listener count
▶	Rujiphart	14
	Kanyanut	7
	Phuriwat	5
	Pongpop	4
	Bunradar	0
	Waranat	0
	Padon	0
	Kanin	0
	Watsapol	0
	Paveena	0

10. Find the user that has never called the operator.

```
SELECT U.USER_ID, U.NAME AS USERNAME
FROM USER U
WHERE U.USER_ID NOT IN(
    SELECT DISTINCT U.USER_ID
    FROM usercalloperator UC
    INNER JOIN user U ON UC.USER_ID = U.USER_ID
);
```

	USER_ID	USERNAME
▶	4	Waranat

11. Count view for each track with user name

```
SELECT u.NAME AS "UserName", t.TRACK_NAME AS "Track name", COUNT(th.TRACK_ID) AS
"listener count"
FROM User u
LEFT JOIN Track t ON u.USER_ID = t.USER_ID
LEFT JOIN TrackHistory th ON t.TRACK_ID = th.TRACK_ID
GROUP BY u.USER_ID, t.TRACK_NAME
ORDER BY COUNT(th.TRACK_ID) DESC;
```

	UserName	Track name	listener count
▶	Rujiphart	Diamond City Lights	6
	Phuriwat	Black Out	5
	Kanyanut	Big Iron	4
	Rujiphart	One	3
	Rujiphart	Last Stardust	3
	Rujiphart	Gracernote	2
	Kanyanut	March of Time	2
	Pongpop	Let It Be	2
	Pongpop	Pretender	2
	Kanyanut	chase	1
	Bunradar	NULL	0
	Waranat	NULL	0
	Padon	NULL	0
	Kanin	NULL	0
	Watsapol	NULL	0
	Paveena	NULL	0

12. Count track that never be in any post

```
SELECT COUNT(*) AS "number of track that never be in post"
FROM Track t
WHERE t.TRACK_ID NOT IN (
    SELECT DISTINCT TRACK_ID
    FROM TrackInPost
);
```

	number of track that never be in post
▶	0

13. Count station's follower, sort by follower

```
SELECT s.STATION_NAME, COUNT(ufs.USER_ID) AS "follower"
FROM UserFollowStation ufs
RIGHT JOIN Station s ON ufs.STATION_ID = s.STATION_ID
GROUP BY ufs.STATION_ID
ORDER BY COUNT(ufs.USER_ID) DESC;
```

	STATION_NAME	follower
►	Lament47	7
	Ki	4
	sora lee	4
	Bookiezz	3
	Jeanne-MuNe	3
	Moggy3.0	3
	Onlikiszz	2
	urata	2
	Vocaloider	1
	Jyu / Just Q	1

14. count repost for each user when its has been repost by other (not oneself), sort by repost count desc

```
SELECT u.NAME, COUNT(urt.REPOST_ID) AS "repost count"
FROM UserRepostTrack urt
RIGHT OUTER JOIN User u ON urt.USER_ID = u.USER_ID
LEFT OUTER JOIN Track t ON urt.TRACK_ID = t.TRACK_ID
WHERE (u.USER_ID != t.USER_ID)
      OR (urt.USER_ID IS NULL)
GROUP BY u.USER_ID
ORDER BY COUNT(urt.REPOST_ID) DESC;
```

	NAME	repost count
►	Paveena	6
	Pongpop	5
	Bunradar	4
	Waranat	4
	Kanyanut	4
	Kanin	3
	Phuriwat	2
	Watsapol	1
	Padon	0

15. Count the numbers of operations in each month.

```

SELECT MONTH(UO.TIMESTAMP_AT_CALL) AS "Month",
       COUNT(MONTH(UO.TIMESTAMP_AT_CALL)) AS NUMBER_OF_OPERATION
FROM usercalloperator UO
INNER JOIN operator O ON UO.OPERATOR_ID = O.OPERATOR_ID
GROUP BY MONTH(UO.TIMESTAMP_AT_CALL)
ORDER BY MONTH(UO.TIMESTAMP_AT_CALL) ASC;

```

Month	NUMBER_OF_OPERATION
1	3
2	1
3	2
5	2
6	4
7	8
8	1
9	4
10	4
11	1

16. Find the most blocked user.

```

SELECT ubu.USER_ID2 AS "UserID", u.NAME AS "UserName", COUNT(ubu.USER_ID1) AS
"Blocked by"
FROM userblockuser ubu
INNER JOIN User u ON ubu.USER_ID2 = u.USER_ID
GROUP BY ubu.USER_ID2
ORDER BY COUNT(ubu.USER_ID1) DESC
LIMIT 1;

```

	UserID	UserName	Blocked by
►	5	Kanyanut	5

17. Show the history of track

```
SELECT HISTORY_ID, User.USER_ID, NAME, TRACK_NAME
FROM TrackHistory
INNER JOIN User ON TrackHistory.USER_ID = User.USER_ID
INNER JOIN Track ON TrackHistory.TRACK_ID = Track.TRACK_ID;
```

	HISTORY_ID	USER_ID	NAME	TRACK_NAME
▶	1	1	Rujiphart	Black Out
	20	1	Rujiphart	Big Iron
	28	1	Rujiphart	Black Out
	7	2	Phuriwat	Black Out
	10	2	Phuriwat	March of Time
	22	2	Phuriwat	One
	15	3	Bunradar	Diamond City Lights
	24	3	Bunradar	Last Stardust
	25	3	Bunradar	Last Stardust
	4	4	Waranat	Pretender
	19	4	Waranat	Big Iron
	23	4	Waranat	Diamond City Lights
	30	4	Waranat	Black Out
	8	5	Kanyanut	One
	9	5	Kanyanut	March of Time
	11	5	Kanyanut	Diamond City Lights
	12	5	Kanyanut	Gracernote
	17	5	Kanyanut	Let It Be
	26	5	Kanyanut	Big Iron
	3	6	Padon	One
	13	6	Padon	Let It Be
	18	6	Padon	Gracernote
	5	7	Pongpop	Big Iron
	14	7	Pongpop	Diamond City Lights
	21	7	Pongpop	chase
	2	8	Kanin	Black Out
	29	8	Kanin	Pretender
	6	9	Watsapol	Diamond City Lights
	16	9	Watsapol	Last Stardust
	27	10	Paveena	Diamond City Lights

18. Show users who have follow the station

```
SELECT Station.STATION_ID, STATION_NAME, NAME, STATION_NAME
FROM UserFollowStation
INNER JOIN User ON UserFollowStation.USER_ID = User.USER_ID
INNER JOIN Station ON UserFollowStation.STATION_ID = Station.STATION_ID
ORDER BY STATION_ID
LIMIT 10;
```

	STATION_ID	STATION_NAME	NAME	STATION_NAME
▶	1	Onlikiszz	Phuriwat	Onlikiszz
	1	Onlikiszz	Bunradar	Onlikiszz
	2	Voccaloider	Waranat	Voccaloider
	3	Bookiezz	Phuriwat	Bookiezz
	3	Bookiezz	Bunradar	Bookiezz
	3	Bookiezz	Watsapol	Bookiezz
	4	Jeanne-MuNe	Rujiphart	Jeanne-MuNe
	4	Jeanne-MuNe	Pongpop	Jeanne-MuNe
	4	Jeanne-MuNe	Kanin	Jeanne-MuNe
	5	Lament47	Rujiphart	Lament47

19. Show the usernames that comment to the track with words “no” or “didn’t” inside the comment.

```
SELECT uct.USER_ID, u.NAME, uct.TRACK_ID, t.TRACK_NAME, uct.COMMENT
FROM usercommentontrack uct
LEFT JOIN user u ON uct.USER_ID = u.USER_ID
LEFT JOIN track t ON uct.TRACK_ID = t.TRACK_ID
WHERE uct.COMMENT LIKE "%didn't%" OR uct.COMMENT LIKE "%no%";
```

USER_ID	NAME	TRACK_ID	TRACK_NAME	COMMENT
6	Padon	1	chase	He said he was not there yesterday, however
8	Kanin	3	Diamond City Lights	Potato wedges probably are not best for relatio...
2	Phuriwat	4	One	Doris enjoyed tapping her nails on the table to ...
10	Paveena	1	chase	Mary plays the piano.
3	Bunradar	4	One	The beach was crowded with snow leopards.
6	Padon	7	Big Iron	The secret code they created made no sense
8	Kanin	10	Pretender	Gary didn't understand why Doug went upstairs...
4	Waranat	6	Last Stardust	There are no heroes in a punk rock band.
10	Paveena	3	Diamond City Lights	Now I need to ponder my existence and ask my...

20. Show users who call to operator

```
SELECT TIMESTAMP_AT_CALL, User.USER_ID, NAME, Operator.OPERATOR_ID,
OPERATOR_NAME
FROM UserCallOperator
INNER JOIN User ON UserCallOperator.USER_ID = User.USER_ID
INNER JOIN Operator ON UserCallOperator.OPERATOR_ID = Operator.OPERATOR_ID
ORDER BY TIMESTAMP_AT_CALL ASC;
```

	TIMESTAMP_AT_CALL	USER_ID	NAME	OPERATOR_ID	OPERATOR_NAME
▶	2021-01-01 12:27:16	6	Padon	1	Akara
	2021-01-18 05:16:27	1	Rujiphart	5	Pililuck
	2021-01-21 02:43:07	9	Watsapol	7	Ittipon
	2021-02-13 07:12:12	10	Paveena	6	Wudhichart
	2021-03-04 07:14:22	5	Kanyanut	9	Jidapa
	2021-03-14 06:35:49	5	Kanyanut	3	Siripen
	2021-05-12 02:34:16	5	Kanyanut	7	Ittipon
	2021-05-16 01:59:20	8	Kanin	9	Jidapa
	2021-06-15 01:47:22	1	Rujiphart	3	Siripen
	2021-06-20 09:35:14	3	Bunradar	7	Ittipon
	2021-06-23 01:20:55	2	Phuriwat	2	Pawita
	2021-06-24 08:00:16	5	Kanyanut	5	Pililuck
	2021-07-03 04:13:26	6	Padon	7	Ittipon
	2021-07-03 12:38:16	8	Kanin	9	Jidapa
	2021-07-04 09:22:02	1	Rujiphart	1	Akara
	2021-07-07 02:53:39	10	Paveena	5	Pililuck
	2021-07-18 05:17:35	5	Kanyanut	1	Akara
	2021-07-23 05:51:11	3	Bunradar	10	Ananta
	2021-07-25 12:24:48	4	Waranat	6	Wudhichart
	2021-07-31 06:53:51	7	Pongpop	9	Jidapa
	2021-08-02 04:47:52	10	Paveena	4	Boonsit
	2021-09-07 08:53:18	5	Kanyanut	3	Siripen
	2021-09-21 04:28:26	2	Phuriwat	9	Jidapa
	2021-09-25 02:04:49	10	Paveena	10	Ananta
	2021-09-26 07:43:39	3	Bunradar	8	Snit
	2021-10-17 01:59:08	8	Kanin	7	Ittipon
	2021-10-18 01:00:19	10	Paveena	10	Ananta
	2021-10-18 07:40:05	10	Paveena	3	Siripen
	2021-10-20 10:32:06	1	Rujiphart	1	Akara
	2021-11-13 04:26:04	10	Paveena	3	Siripen

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